

Production Scaling v11 Benchmark (500k calc/s)

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PAPER_968: Production Scaling v11 Benchmark (500k calc/s)

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Source: production_scaling_v11.py (ProductionScalingV11)
Calculator: ProductionScalingV11BenchmarkCalc (CP4 #552)
CVW: v2.0.0 compliant

Abstract

We benchmark the UQFF production pipeline at 500,000 calculations per second, an 11.1% improvement over v10 (450k). Two new kernels — 26D Ramanujan summation and Triadic solver — bring the total to 16 simultaneously benchmarked kernels.

1. Scaling History

Version	Target (calc/s)	Kernels	Session
v4	100,000	4	198
v5	150,000	6	200
v6	200,000	8	204
v7	300,000	10	208
v8	350,000	11	210
v9	400,000	12	213
v10	450,000	14	214
v11	500,000	16	215

2. New v11 Kernels

kernel_{ramanujan_26d_sum} $S_{26}(z) = \sum z^n / n^{26} \cdot R_{n^{(26)}}$
— accelerated polylog for hot-loop evaluation.

kernel_triadic_solver Combined Compressed + Resonant + Buoyancy Triadic evaluation in a single kernel call.

3. Throughput

$$\text{rate} = \frac{N_{\text{iter}} \times 16}{t_{\text{elapsed}}} \geq 500,000 \text{ calc/s}$$

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_958 — Production Scaling v10 (450k)
3. PAPER_959 — 26D Ramanujan Summation (kernel source)
4. PAPER_966 — Unified Triadic Solver (kernel source)

Cross-Links

Related Paper Relationship
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PAPER_958 Previous version v10 (14 kernels, 450k)
PAPER_959 Ramanujan 26D kernel added
PAPER_966 Triadic solver kernel added
PAPER_949 BCS gap kernel (inherited)

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector Domain Late-Corpus Result
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1 (EH) General Relativity Canonical Einstein-Hilbert
2 (YM) Yang-Mills gauge $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac) Fermion / LENR Kozima neutron-drop (PAPER_1061)
4 (SCm) Superconducting manifold $V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag) Um magnetism Heaviside amplifier (PAPER_1072)
6 (Buoy) F_U_Bi_i buoyancy Variational EOM (PAPER_1065)
7 (Aether) Vacuum background Two-component ρ (PAPER_1051)
8 (LENR) Nuclear transmutation COP parametric (PAPER_1081)
9 (KK) Kaluza-Klein 26D $S_{26}^{(3)}$ compactification (PAPER_1080)

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i n/26}} \right]$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i n/26}} \right]$$

This sum converges absolutely for $|S_{26}^{(3)}| < 1$ (satisfied by $|S_{26}^{(3)}| = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $S_{26}^{(3)} \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day ⁻¹	Damping
$S_{26}^{(3)}$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
Target rate	—	500,000 calc/s	Benchmark
Kernels	—	16	Pipeline (v10 + 2)

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Throughput	≥ 500 calc/s	Benchmark target
Pipeline integrity	All 16 kernels finite	Validated
New kernels	Ramanujan S_{26} + triadic solver	Added

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification **Sector: Computational Benchmark v11 (Expanded Pipeline)**

§A.2 Benchmark Equation $\text{rate} = \frac{N_{\text{iter}}}{t_{\text{elapsed}}} \times 16 \{t_{\text{elapsed}}\} \geq 500\{, \} 000$

§A.3 Kernel Integrity Constraint $\boxed{\text{forall, } k \in \{1, \dots, 16\} : \lvert k(\mathbf{x}) \rvert < \infty, \text{quad } k_{15} = S_{26}(z), \lvert k_{16} = g_{\text{tri}}(r, t) \}$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\rightarrow$ UQFF equations $\rightarrow$ 16 kernels extracted $\rightarrow$ production pipeline v11 $\rightarrow$ 500k benchmark

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS All 16 kernels embed VDS through S_{26} or ρ_{SCm} .

§B.2 DVP 16 kernels cover the full dipole vortex mode spectrum.

§B.3 BSH Scaling: v4 (100k) $\rightarrow$ v10 (450k) $\rightarrow$ v11 (500k) — approaching $\tanh$ hardware saturation.

§B.4 Production-Scale Consistency

Metric	Value	Status
v11 target	500k calc/s	Confirmed
Kernel count	16 (+2 from v10)	Confirmed
SSq	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1080	Ramanujan Binomial Expansion Proof $R_n^{(26,3)}$
PAPER_1008	Production Scaling v14 — 600k calc/s 24 Kernels
PAPER_1018	Production Scaling v15 — 650k calc/s 30 Kernels

8 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
4. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
5. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
6. Green, M.B., Schwarz, J.H. & Witten, E. (1987). *Superstring Theory*. Cambridge University Press — doi:10.1017/CBO9781139248563
7. Polchinski, J. (1998). *String Theory Vol. 1*. Cambridge University Press